Patient Specific Stereotaxic Guide Specifications for Dental Germinative Photodynamic Regulation

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Project Scope:

This document covers design specifications for an alpha-stage dry-bone prototype of a novel stereotaxic surgical guide, designated the Auriel Stereotaxic Guide. The system is engineered to accommodate the precision requirements of Dental Germinative Photodynamic Regulation (dGPR), which is a theoretical therapeutic protocol designed to arrest retrograde root elongation using targeted photodynamic therapy for stem cell apoptosis.

The clinical goal of this frame is to facilitate minimally invasive access to root apices without the need for traumatic extractions, with the extreme precision required to isolate targets like the root apices.

Data Source: *Chinchilla lanigera.* cranium and mandible models sourced from the Natural History Museum of Los Angeles County via MorphoSource.org (https://www.morphosource.org/concern/media/000114253)

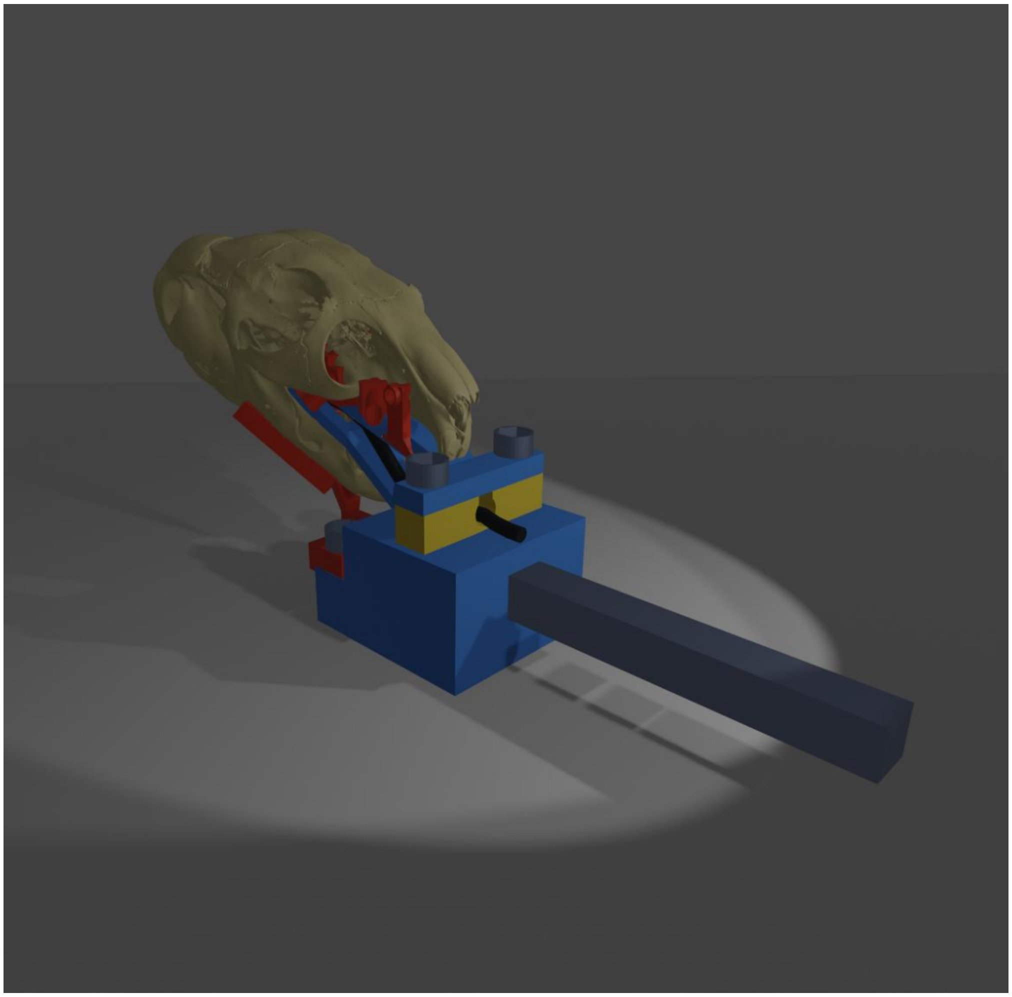


Figure 1

Prototype Status

To reiterate, the current iteration is a prototype constructed with simplified geometry as a functional proof of concept, modelled on dry skeletal data. In future stages, intended for cadaver studies or live patients, edges (particularly those that touch soft tissue) will be rounded to prevent mucosal injury. Additionally, future models will have a surface offset in areas with soft tissue (such as the skin and muscle around the outer surface of the mandible). Certain elements, particularly the external mandibular blocks, are depicted in reduced forms for clarity, a clinical version will have additional reinforcement to maximize rigidity under surgical load.

Device Components

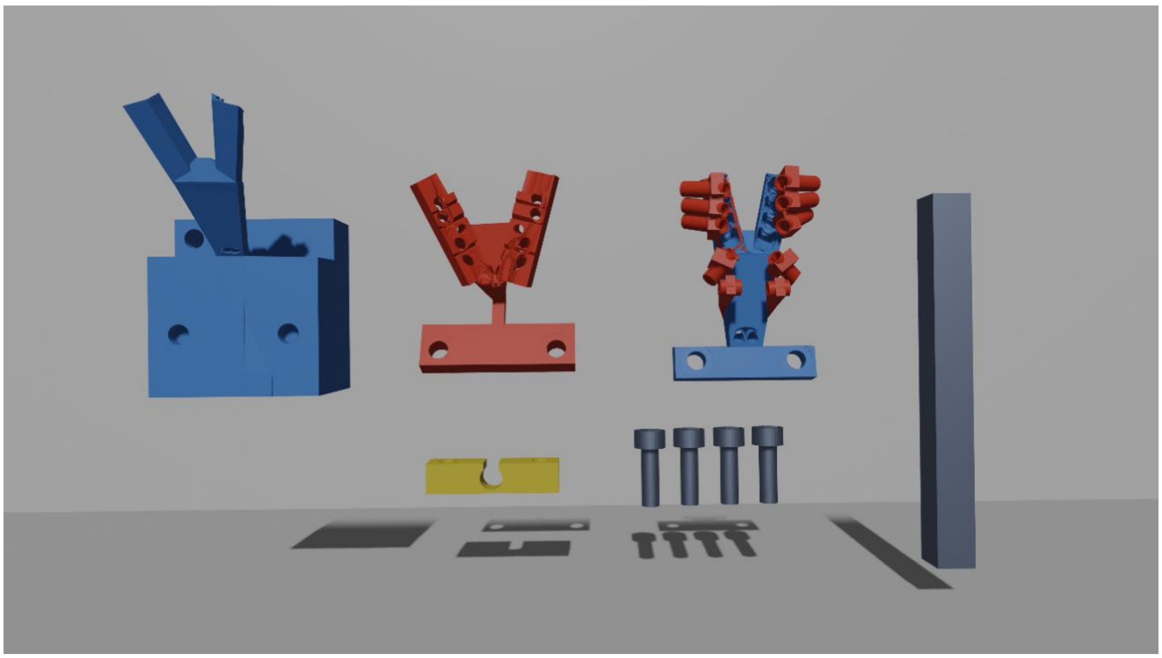


Figure 2

Figure 2 displays the components of the device. The stereotaxic guide is composed of four primary pieces (after the metal rod is mounted, which can be done with surgically appropriate adhesives).

Note: the maxillary barrels are depicted with cylindrical ends for visual clarity, these would have to be refined before fabrication using a Boolean difference operation (with a 1mm offset to account for shrinkage of print material) to produce a surface that correctly conforms to the patient's maxilla.

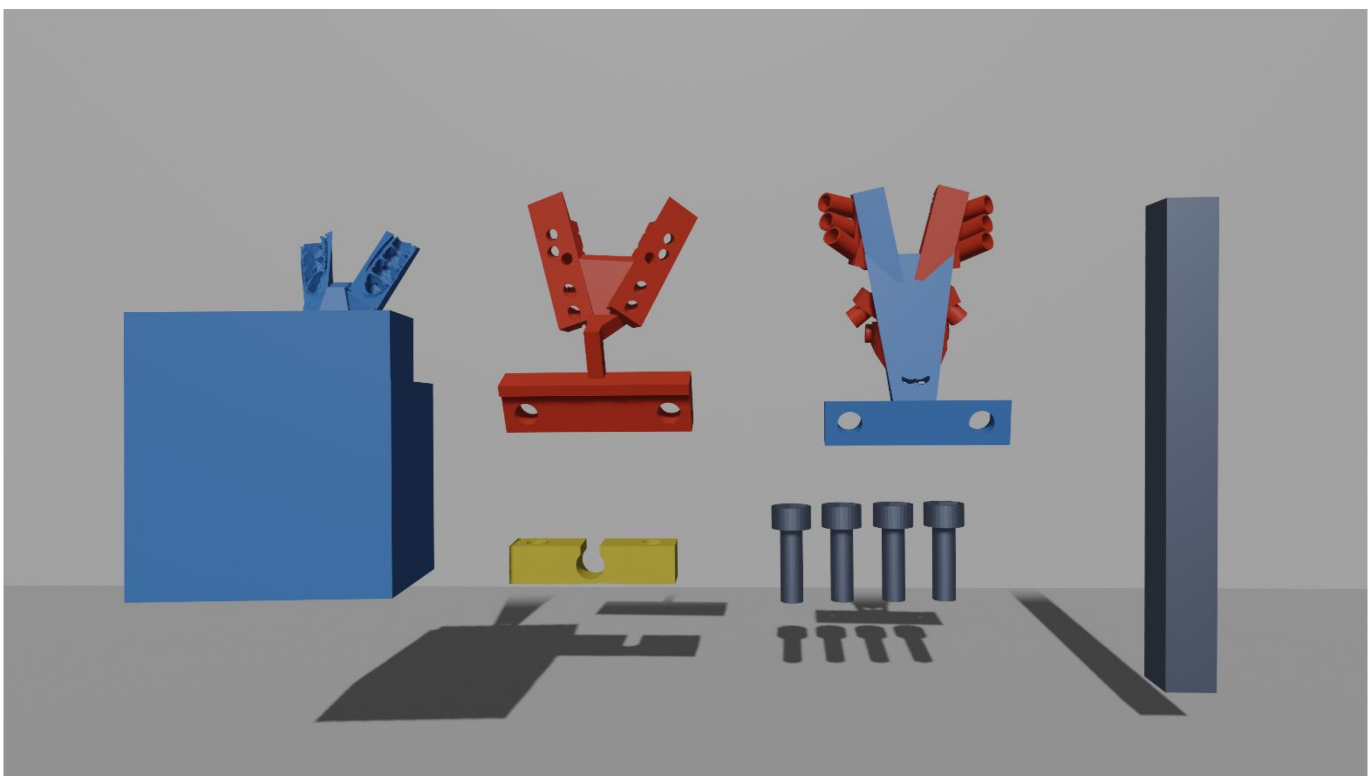
The base of the guide (leftmost, solid blue) is the housing for the metal rod, which also makes it the mounting point between the system and a surgical table clamp. The base also functions as the intra-oral fixation point, providing a direct link between the surgical clamp and the chinchilla's mandible.

The intubation guide block (bottom-left, yellow), allows the intubation tube through the stereotaxic fixation device, with a keyhole design to allow it to be affixed after intubation. It was additionally designed with significant clearance to enable rapid access to the airway and emergency extubation when necessary.

The external mandibular blocks (second from the left, red) guide the surgeon's drill down a precalculated path to the germinative center of the chinchilla's tooth. The barrels should be fitted with surgical-grade steel inlays prior to use in order to prevent the drill from damaging the resin and introducing fragments into the chinchilla's mandible. They have a separation from the bone at the masseter's insertion to prevent damage or compression of the facial artery and vein, given its position aligns with related species (Greene, 1935; Cooper, 1975; Natural History Museum of Los Angeles County, 2020).

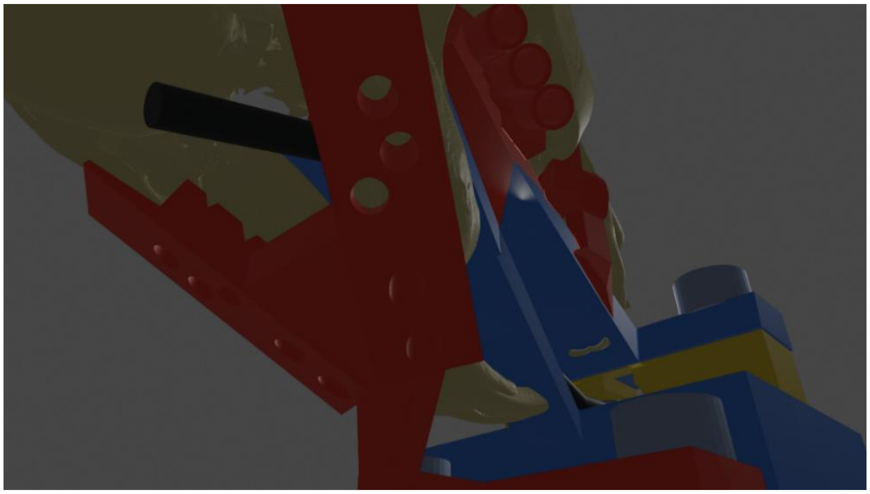
The maxillary block (third from the left) is a fixation device with attached drill guides. The fixation segment conforms to the teeth to form a solid lock. The barrels are built onto the fixation device (their current design would require reinforcement prior to use) and should be fitted with a steel lining in order to prevent damage to the barrels from the drill. The guides on the maxillary block were designed for palpability, so the surgeon can feel where they need to cut for the most direct path to the germinative centers.

Lastly, the device uses four 10mm M3 screws for structural stability.

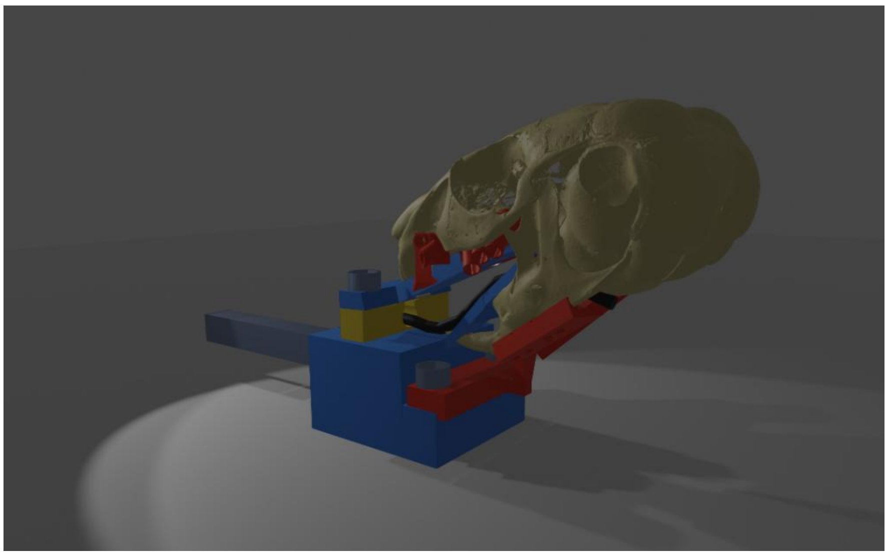


Clinical Justification for Patient-Specific Design

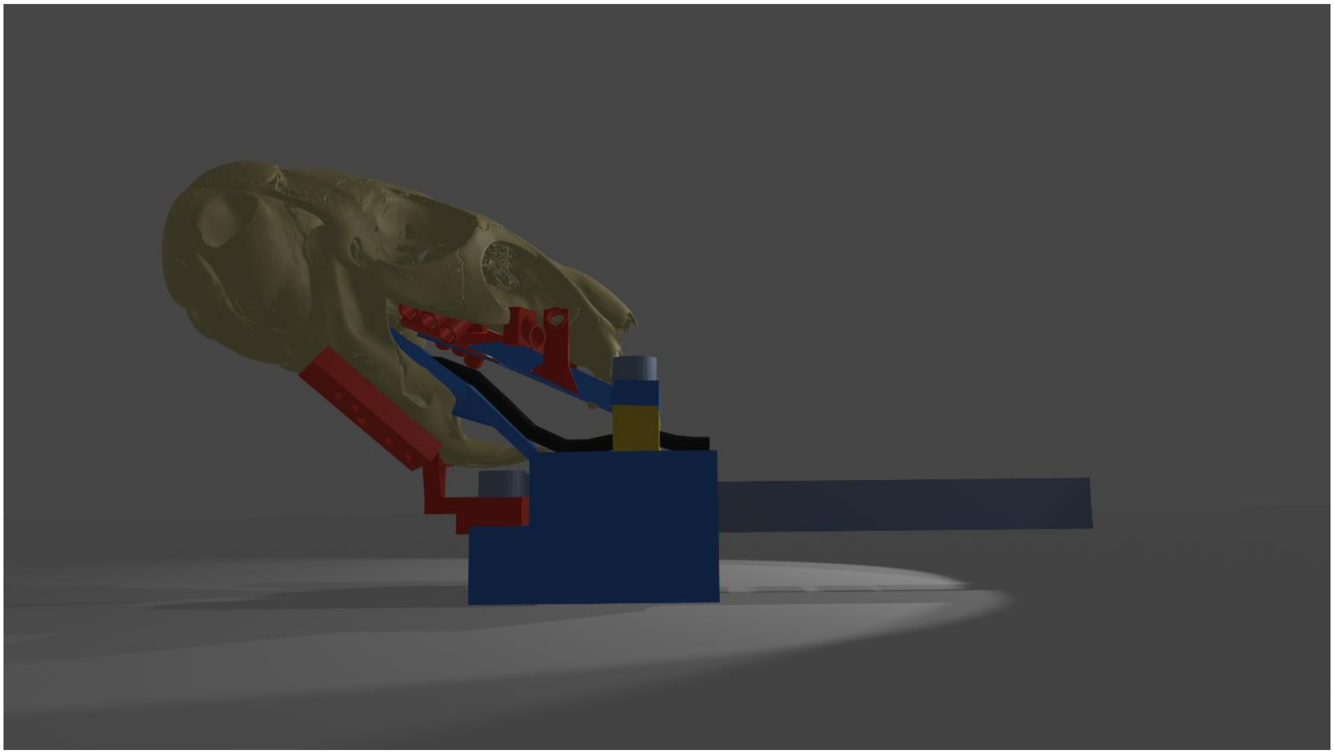
For this conceptual design a relatively typical chinchilla scan was used, and the stereotaxic guide was built around the bone. Even in this relatively typical chinchilla model, the germinative centers were misaligned between the two sides of the mandible, this reinforces the need for patient-specific models, as even healthy chinchilla roots are not always static in one location.



The above image demonstrates that the external mandibular blocks have triangular grooves running perpendicularly to the jaw, which are intended to prevent saline and blood accumulation against the skin. Additionally, there is a large gap between these blocks and the mandible as they approach the groove created by the masseter's insertion, where the facial artery and vein are assumed to curve around the jaw to supply blood to the face due to the vascular anatomy of related species (Greene, 1935; Cooper, 1975; Natural History Museum of Los Angeles County, 2020), a detail vital to the welfare of the patient, which due to the lack of primary data, will have to be verified in the cadaver trial phase, such that adjustments can be made if necessary.



The external mandibular blocks could not be directly affixed to the primary structure alongside the intra-oral mandibular block, as it would make fixation of the device functionally impossible, and likely to cause structural damage which would make the guide system inaccurate. As such, the external mandible blocks are affixed to the main structure after it is inserted into the oral cavity. It is affixed using an L-shelf design that provides simplicity in preparation, ease of use, and prevents rotational motion of the blocks during surgery.



Some features for future designs are already in the planning phase, including the insertion of ports into drill barrels for ease of saline cooling; attachable depth-stops which physically prevent a drill from extending into the soft tissue; and an external maxillary guide that allows for treatment of maxillary cheek teeth that have undergone retrograde elongation, through a trans-orbital approach, using specialized spatulated guides with steel inlays to shield the globe from light and the drill, safely retracting and protecting the globe while accessing the alveolar floor.

Notably, I believe a generalist version of this model (i.e. a version without the drill barrels and extra-mandibular guide) would be a valuable operative tool for surgeries throughout the cranium and mandible due to its stability.